Y1S1

Introduction to Programming - Dev C++

Introduction to C Programming	• Basic syntax and semantics of C language • Standard Input and Output • Preprocessor directives
Types, Operators and Expressions	• Variables and primitive data types • constants • Operators, Expressions and assignments • Relational and Logical Operators • Type conversions • Increment and Decrement Operators • Precedence and order of Evaluation
Control Structures	• Statements and Blocks • If-Else • Else-IF • Switch • Loops – While, For and Do –while • break and continue statements • Nested Loops
Functions	• Basics of Functions • Parameter Passing • Avoiding duplication of Code • Recursion
Pointers	• Pointers and Addresses • Pointers and Function arguments • Dynamic Memory Allocation
Arrays	• One – dimensional Arrays • Multi-dimensional Arrays String Handling
Structures	• Basics of Structures
File Processing	• Read text files • Write to text files • Structures and files
Programming practices	• Compiling multiple programs using a make utility • Debugging Strategies to find bugs • Using proper coding standards • Documenting Code Test Driven Development, Unit Testing

Introduction to Computer Systems - Multimedia Logic (MM Logic) software , Cisco Packet Tracer

Data Types and Representation	• Introduction to Data and information • How to store the data in a computer? • Basic Data Types • Number Representation • Floating Point Representation • Introduction to computer architecture • Components of the Computer • Different usages of the computers • Von Neumann Architecture • Different data representation methods in Computers
Logic Simplification-Karnaugh Maps	• Introduction to Integrated Circuits • Characteristics of combinational Circuits • Introduction to Kmaps • History and importance of Kmaps • Rules for simplifying the Boolean expressions using Kmap • Kmap simplification for two, three and four variables • Don't care condition
Combinational and Sequential Logic Circuits	• Combinational and Sequential Circuits • Adders and Shifters • Encoders and Decoders • Multiplexers and De-multiplexers • Programmable Logic Arrays • Sequential Circuits • Latches, Flip-flops and Excitation Tables
Computer Architecture	• The basic structure of a computer • Main components • Memory Hierarchy • Registers • Cache Memory • Main Memory • Hard Disk Drive
Operating Systems	• Definition and Need of OS • Services of the OS • Classification of the OS • Different types of OS
Living in a Network-Centric World	• Communication- An essential part of human lives • Define a computer network • Explain the elements of communication • List and describe the types of networks • The Internet architecture • Trends in networking
Communicating Over the Network	• Explain how data is transmitted and received • List and describe the types of transmission media • Describe the functions of a network interface card

	• List and describe the characteristics different network topologies • Selecting a network topology • Using layered model
ISO/OSI Layered Architecture	• OSI reference model • Define protocol • List and describe the functions of TCP/IP Protocol Model • Explain the functions of client server architecture
Addressing	• Purpose of different types of addresses • Address types (logical, physical & port address) • Logical addressing (IPv4) • Physical Address (MAC Address) • Assigning and Calculating an IP address
Ethernet	• Overview of Ethernet • Use of Ethernet for communication through a LAN • Ethernet frame (HDLC) • Ethernet media access control methods • Ethernet physical layer • Address resolution protocol
Design a simple Network	• Identify Basic Network Devices and their interfaces. • Media Selection for the Communication • Device interconnections • Address scheme development

Mathematics for Computing - GNU Octave

Logic Control-Part I	• Introduction to Boolean Algebra o Laws of Boolean Algebra, Boolean Algebra Rules • Simplifying Boolean Expressions using Boolean Algebra • Logic circuits design process (SOP) • Deriving logical expressions (SOP)
Logic Control-Part II	• Introduction to Logic Gates o Basic Logic Gates, Derived Logic Gates, Special Logic Gates • Truth Tables for Gates • Universality of NAND & NOR Gates • Predicates • Logic of quantified Statements
Number Systems	• Introduction to Number bases • Number Base Conversions • Conversion of Fractions Computer Arithmetic • Introduction to Computer Arithmetic o Binary Addition, Subtraction, Multiplication and division • Sign Magnitude Representation • Floating Point Numbers
Introduction to differentiation	• Tangent line to a graph • Slope of the graph

	• The derivative of a function • Some rules of differentiation • The product rule • The quotient rule • The chain rule
Integration and its Applications	• Antiderivatives and indefinite integrals, • Definition of antiderivative • Basic integration rules • Area and the fundamental theorem of calculus
Functions	• Equality of functions • Identity function • One-to-one functions • Onto functions • One-to-one correspondence • Inverse function • Pigeon-hole principle
Counting	• Sequences • Finding Sum of a Series • Properties of Sequences • Permutations • Combinations • Binomial Theorem
Graph Theory	• Basic definitions • Degree of a vertex • Paths • Circuits • Cycles and Connectivity
Graph Theory continued	• Digraphs and relationship graphs • Isomorphism of graphs • Euler and Hamiltonian paths
Matrices	• Matrix definition and notation • Rules of matrix algebra • Special matrices • Determinants The determinant of a square matrix
Matrices II	• The adjoint and inverse matrices • Inverse matrix with the Gaussian elimination Solving Linear equations • Solution of linear equations by Gaussian elimination • Solution of linear equations by Cramer's Rule • Compatible and incompatible sets of equations
Finite-state machines	• Definitions • Representing a Finite State Machine using a Transition Diagram or an annotated next-state table • Language accepted by a machine

Y1S1

Object-Oriented Concepts - Visual Studio

Introduction to C++	• Introduction to C++ • Basic programing with C++ • cout and cin commands • Using Namespaces • How to use the Visual Studio Integrated Development Environment
Introduction to OOP Concepts	• Introduction to Object Oriented Concepts • classes, objects • properties and Methods
Introduction to Object Oriented Programming	• Introduction to Object Oriented Programming • Constructors / Destructors • Encapsulation • Information/Implementation Hiding • Reference type parameters • Function Overloading
Identifying classes and objects	• Identifying classes and objects using • Noun and verb analysis • CRC Cards • Abstraction
Object Oriented Design	Object Oriented Design • Interfaces • Dependency, Association, Aggregation, Composition • Inheritance, Polymorphism
Advanced Object Oriented Concepts	• Inheritance, Polymorphism, virtual functions • Implementing class relationships Association, Aggregation and Composition • Static Properties and Methods

Information Systems and Data Modeling – Microsoft SQL Server Management Studio

Information systems in business organizations	• Introduction to Information System (IS) • The role of ISs in business • Types of IS used in business • Information systems users • Improving business using Information Systems • Benefits of using information systems in business
Information systems for achieving organizational strategies	• Decision making and Information Systems • Strategic business objectives of Information Systems • Decision support for operational, tactical and strategic management

Competitive advantages in using information systems	• Implications for the design and understanding of Information Systems • The role of an IS professional in developing Information Systems • Sustaining competitive advantage in using IS • Performing a strategic systems analysis in using IS
Modern Information Systems in Business	• Introduction to electronic commerce
Conceptual and Logical Database Design	• Introduction to requirement analysis and modelling • Components of the ER model • ER diagrams • Relational Model • ER to Relational model mapping
Schema Refinement	• Anomalies in a schema • Keys and functional dependencies • Normal forms (1NF, 2NF, 3NF) • Normalization and de-normalization process
SQL DDL	• Creating database tables • Defining integrity constraints
SQL DML and Extensions of SQL	• Using 'SELECT' clause with options • Using aggregate functions • Joining tables • Nested queries • If statements • Case statements • Functions

Software Process Modeling

Introduction to Software Engineering	• The software engineering concepts
Life Cycle Modeling	• Discussion of software process models
Requirements Engineering	• Types of requirements • Eliciting requirements • Handling imprecise requirement • Use Case diagrams and scenarios • Activity diagrams
Software Design	• Levels of design • Design principles • Design strategies • Modelling design
Implementation and Testing	• Implementation • Types of testing • Levels of testing • Writing test cases
Modern Software Development Methodologies	• Agile development methodologies • SCRUM roles, processes, artifacts.

Internet and Web Technologies - text editors (notepad or notepad++), Internet Browser (Internet Explorer or Google Chrome)

Concepts and technologies associated with the Internet applications.	• Internet addressing and protocols • DNS resolution • Data across the internet
Social Media	• Introduction, examples, Pros & Cons of social media • Effective use of social media. • Social media marketing and analytics
e-Commerce and other Web based systems	• Different types of e-commerce • The process of E-commerce and online payment handling • Advantages and Disadvantages of E-Commerce
Implementation Technologies (HTML/CSS)	Implementation Technologies (HTML/CSS) • Introduction to HTML/CSS • HTML / CSS validation • Best practices and coding standards
Implementation Technologies (JavaScript)	• Introduction and basics of JavaScript • JavaScript functions, event and form handling • Best practices and coding standards • JavaScript testing
Implementation Technologies (PHP)	• Introduction and basics of PHP • Functions and arrays • Form processing and validation • Best practices/coding standards
Web standards, Usability and Accessibility	• Web standards • Usability guidelines • Web Accessibility Initiative (WAI) • Guidelines for Multimedia on the Web
Information security and privacy on the Internet	• Internet security threats, services and attacks • The world of Cryptology • e-Life, privacy and security

Y2S1

Software Engineering - Star UML

Object Diagram
Sequence Diagram
Communication Diagram
State Machine Diagram
Component Diagram
Physical Diagram
Deployment Diagram
Software Testing
Design Pattern

Object Oriented Programming – Eclipse ,tomcat

Class & Object
Static Members
Abstract classes and Interface
Error and Exceptions
String
Collection Framework
Generics
Threads
Design Patterns

Database Management Systems - Microsoft SQL Server Management Studio , MySQl

Requirement Analysis	• Data requirements • User identification • Security requirements • Performance requirement
Conceptual Design	• EER model (ISA hierarchies and aggregation) • Design traps
Logical Database Design	• Relational model • ER to relational mapping • ISA hierarchy and aggregation mapping
Schema Refinement	• Use of Armstrong's Axioms to find closure for set of functional dependencies • Use of Attribute closure to find functional dependencies and keys • Normal Forms
Query Languages	• Formal query languages • Advanced SQL
Database Programming	• T-SQL programming constructs • Functions • Stored procedures • Views • Triggers
Database Connectivity	• Open Database Connectivity (ODBC) Architecture • Java Database Connectivity (JDBC) Architecture • Type of JDBC drivers • JDBC classes and interfaces • Prepared statements • Exception handling • Transaction handling • Calling functions and stored procedures
Database installation, configuration and data migration	• Installation and configuration of SQL server • SQL Server Integration Services (SSIS) • Bulk copy program (bcp)
Database Maintenance	• Jobs creation • Job scheduling • Database backups and restoring

Database Security	• Database server authentication methods • Server and database Roles • Ownership and user-schema separation • Authorization and permissions • Data encryption
Database Performance	• Performance monitoring • Analysis of query execution • Selection and creation of indexes

Computer Network – Cisco Packet Tracer

Introduction to Router
Ip Address and VLSM
IPV6
Routing Concept
IP
Switch
Switch Port Security
STP
TCP
ACL

Operating Systems and System Administration – Ubuntu

Introduction to OS	• Definition, Purpose, Development, and Design of OS • OS structures – System components, OS services, system calls, system programs, system structure, system design and implementation • Overview of Computer System structures.
Processes and Threads	• Process and Thread Concept • Process and Threads scheduling, Process operation and cooperation • Inter-process communication.
Process synchronization	• Concept of process synchronization • Semaphores • Classical problems of synchronization • Monitors
CPU Scheduling	CPU scheduling: Basic concepts, CPU scheduling criteria Scheduling Algorithms Algorithms evaluation.
Deadlocks	• Deadlocks Model • Characterization • Deadlock handling methods
Memory Management	• Memory Management Background • Address space • Swapping • Contiguous allocation

	• Paging
Virtual Memory	• Virtual memory Background • Demand Paging and its performance • Page replacement algorithms • Thrashing
File System	• File System Interface and Implementation • File concept • Access Methods • Protection • Implementation
I/O Devices and Secondary Storage	• I/O Systems • Mass-Storage • I/O hardware • I/O interface • Disk structure • Disk scheduling
Protection and Security	• Protection and Security – Goals • Access matrix • Security Problem • Program and System Threats

Y2S2

Information Technology Project – VS code , mongodb,github

Introduction to Information Technology Project
Introduction to Agile Software Development
User Interface Design concepts
ER Evaluation
Software Testing
Document writing

Probability & Statistics – R Studio

Introduction to Statistics
Sampling Methods
Descriptive Statistics
Probability
Random variables & Probability Distributions
Continuous Probability Distributions
Sampling Distributions
Statistical Inference
Chi squared Tests
Regression Analysis

Data Structures and Algorithms - Eclipse , IDLE

Introduction
Stack data structures
Queue data structures
Linked List data structures
Tree data structures
Introduction to Algorithms and Asymptotic Notations
Algorithm designing techniques
Introduction to Heaps
Introduction to searching and sorting algorithms

Employability Skills Development Seminar

Importance of Soft Skills
Expectations of the Industry
Job roles for IT Graduates
Different tools and technologies used in the industry.
Ethics and Professionalism

Mobile Application Development - Figma , Android Studio

Introduction to App Development
Mobile Platforms and Application Development fundamentals
Mobile Interface Design Concepts and UI/UX Design
Introduction to Android Operating System
Main Components of Android Application
Sensors and Media Handling in Android Applications
Data Handling in Android Applications
Android Application Testing and security aspects